



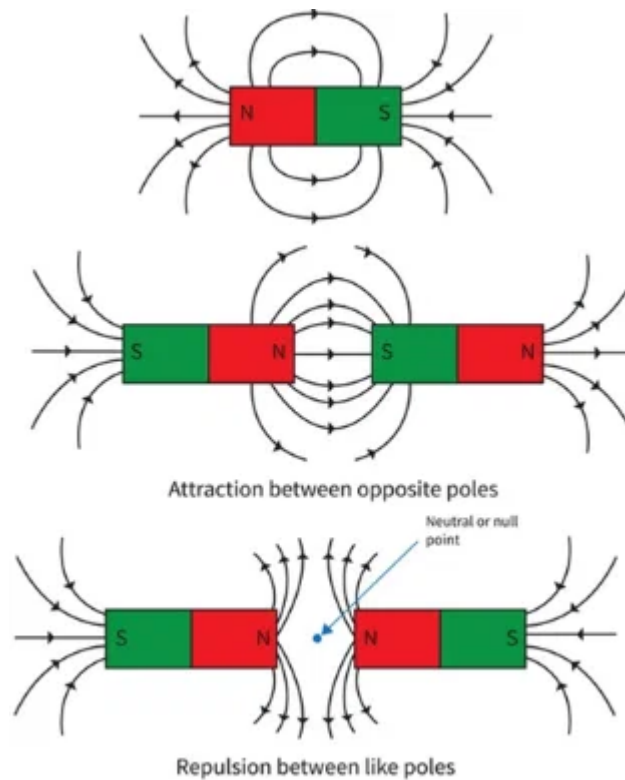
Earth's Magnetic Field, Dynamo theory, Magnetosphere

[🏠](#) > [GS1](#) > [Geography Static](#)

Table of contents

- A 'field' is a region in which a body experiences a force owing to the presence of other bodies. Earth's Magnetic Field is one such field.
- Gravitational fields determine how bodies with mass are attracted to each other.
- In electric fields, objects that have an electric charge are attracted or repelled from each other.
- Magnetic fields determine how electric currents that contain **moving electric charges** exert a force on other electric currents.





Field lines or magnetic flux in a simple bar magnet

Dynamo theory: Generation of Earth's Magnetic Field and Sustaining it

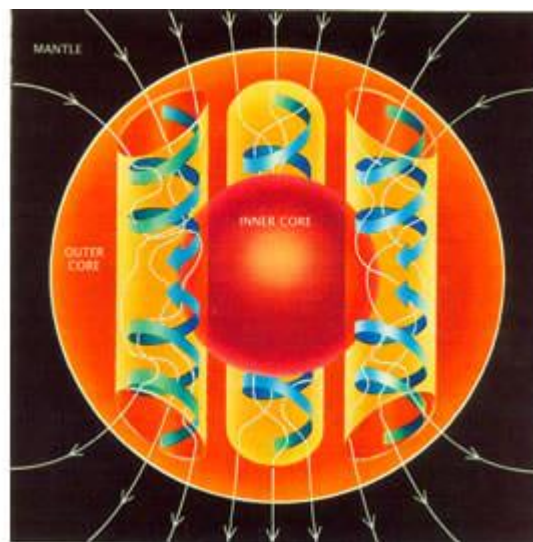
- **Dynamo theory** proposes a mechanism by which a celestial body such as Earth or a star generates a magnetic field and sustains it over astronomical time scales (millions of years).
- Dynamo theory suggests that **convection in the outer core, combined with the Coriolis effect** (caused due to the rotation of the earth), gives rise to **self-sustaining (geodynamo)** Earth's magnetic field.

Mechanism

- Earth's magnetic field is generated in the **earth's outer core**.
- Lower pressure than the inner core means the metal in the outer core is **fluid**.
- The temperature of the outer core ranges from 4400 °C in the outer regions to 6000 °C near the inner core.
- Heat sources include energy released by the compression of the core.



- The differences in temperature, pressure and composition within the outer core cause **convection currents** in the molten iron of the outer core as cool, dense matter sinks while warm, less dense matter rises.
- This **flow of liquid iron** generates **electric currents**, which in turn produce **magnetic fields**.
- Charged metals passing through these fields go on to create electric currents of their own, and so the cycle continues. This **self-sustaining loop** is known as the **geodynamo**.
- The **spiral movement** of the charged particles caused by the **Coriolis force** means that separate magnetic fields created are roughly aligned in the same direction, their combined effect adding up to produce one vast magnetic field of the planet.

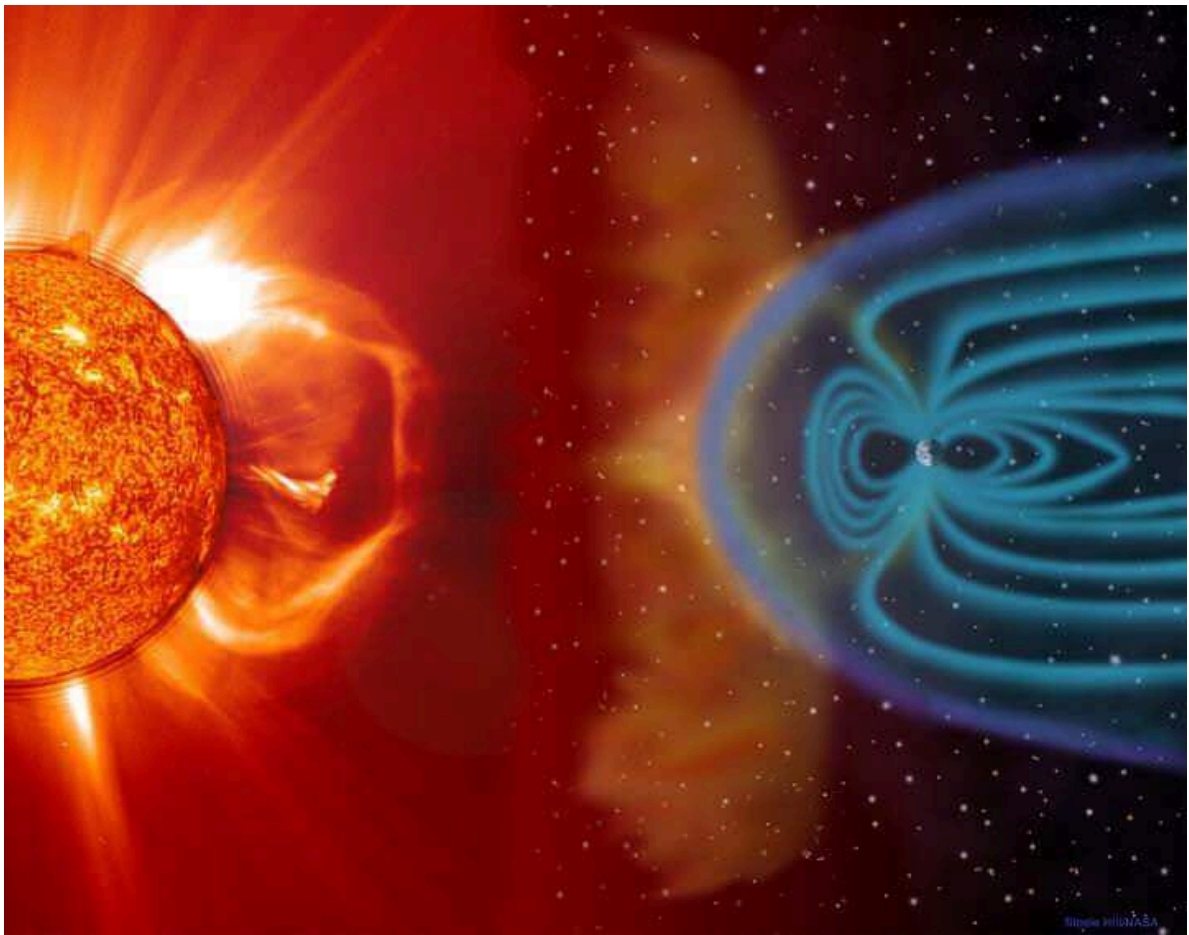


Convection currents in the outer core. Spiral motion is caused due to the Coriolis Effect. ([Wikipedia](#))

Magnetosphere

- The magnetosphere is the region above the ionosphere that is **defined by the extent of the Earth's magnetic field in space**.
- It extends several tens of thousands of kilometres into space, **protecting the Earth from the charged particles of the solar wind and cosmic rays** that would otherwise strip away the upper atmosphere, **including the ozone layer** that protects the Earth from harmful ultraviolet radiation.





Earth's magnetosphere (Credits: losangeles.af.mil)

Magnetopause

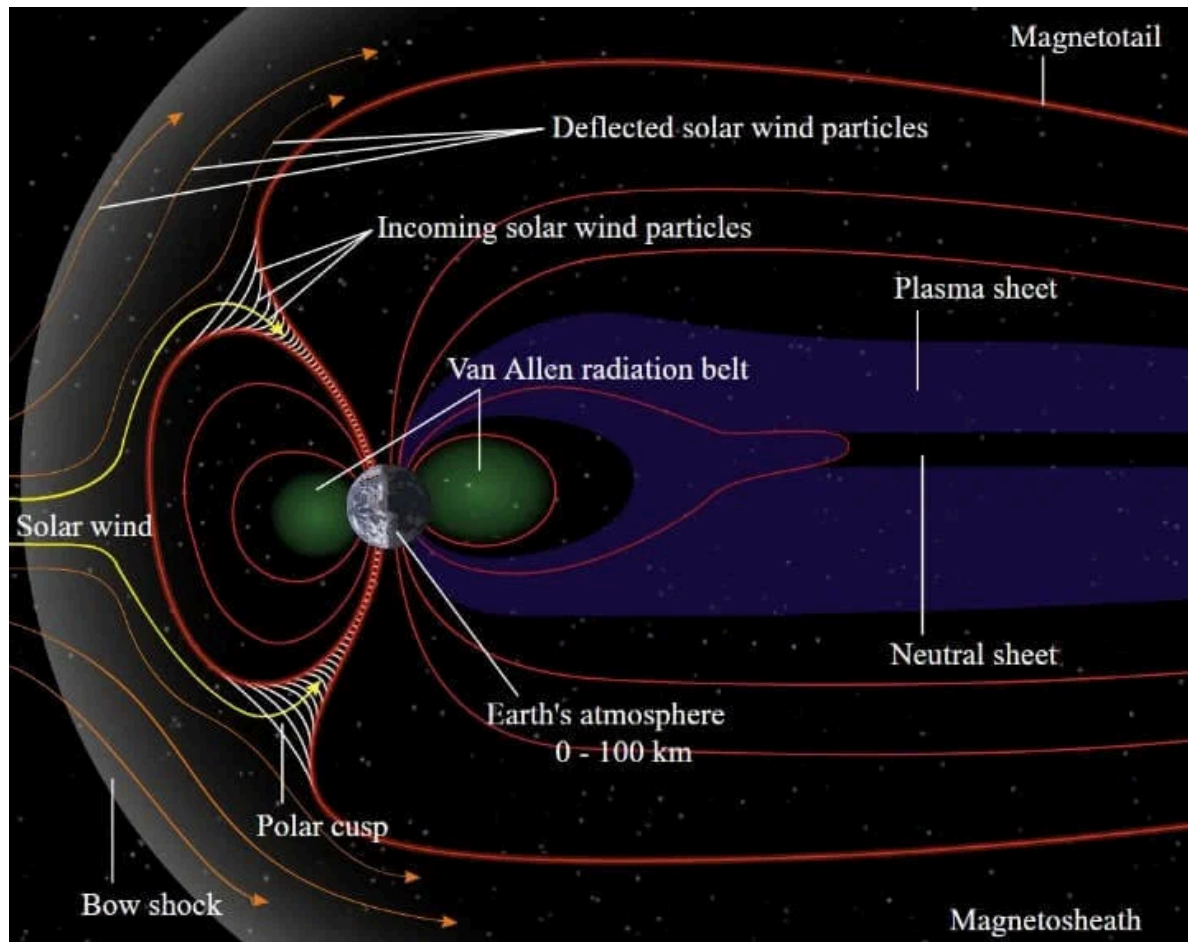
- Earth's magnetic field, predominantly dipolar at its surface, is distorted further out by the **solar wind**.
- The solar wind exerts a pressure. However, it is kept away by the pressure of the Earth's magnetic field.
- The **magnetopause**, the **area where the pressures balance**, is the boundary of the magnetosphere.
- Despite its name, the magnetosphere is asymmetric, with the sunward side being about 10 Earth radii out but the other side stretching out in a magnetotail that extends beyond 200 Earth radii.

Magnetosheath

- The turbulent magnetic region just outside the magnetopause is known as the magnetosheath.



- Sunward of the magnetopause is the **bow shock**, the area where the **solar wind** slows abruptly.



Earth's magnetosphere (Original bitmap from NASA. SVG rendering by Aaron Kaase, via [Wikimedia Commons](#))

Plasmasphere

- Inside the magnetosphere is the plasmasphere, a region containing **low-energy charged particles**.
- This region begins at the height of 60 km, extends up to 3 or 4 Earth radii, and includes the **ionosphere**.
- This region rotates with the Earth.

Auroras

- Aurora is the name given to the luminous glow in the upper atmosphere of the Earth which is produced by charged particles (solar wind)



- Positive ions slowly drift westward, and negative ions drift eastward, giving rise to a **ring current**. This current **reduces the magnetic field at the Earth's surface**.
- Some of these particles penetrate the ionosphere and collide with the atoms there.
- This results in an **excitation of the oxygen and nitrogen molecular electrons**. The molecules get back to their original state by emitting photons of light which are the aurorae.
- The charged particles follow magnetic field lines which are oriented in and out of our planet and its atmosphere near the magnetic poles. Therefore, aurorae mostly are seen to occur at **high latitudes**.



Geomagnetic storms

- The varying conditions in the magnetosphere, known as space weather, are largely driven by **solar activity**.
- If the solar wind is weak, the magnetosphere expands; while if it is strong, it compresses the magnetosphere and more of it gets in.
- Periods of intense activity, called geomagnetic storms, can occur when a **coronal mass ejection erupts** above the Sun and sends a shock wave through the Solar System. It takes just two days to reach the Earth.
- At the Earth's surface, a magnetic storm is seen as a **rapid drop in the Earth's magnetic field strength**.
- **Ring Current**: Ring current is the name given to the large electric current that circles the Earth above its equator during magnetic storms.

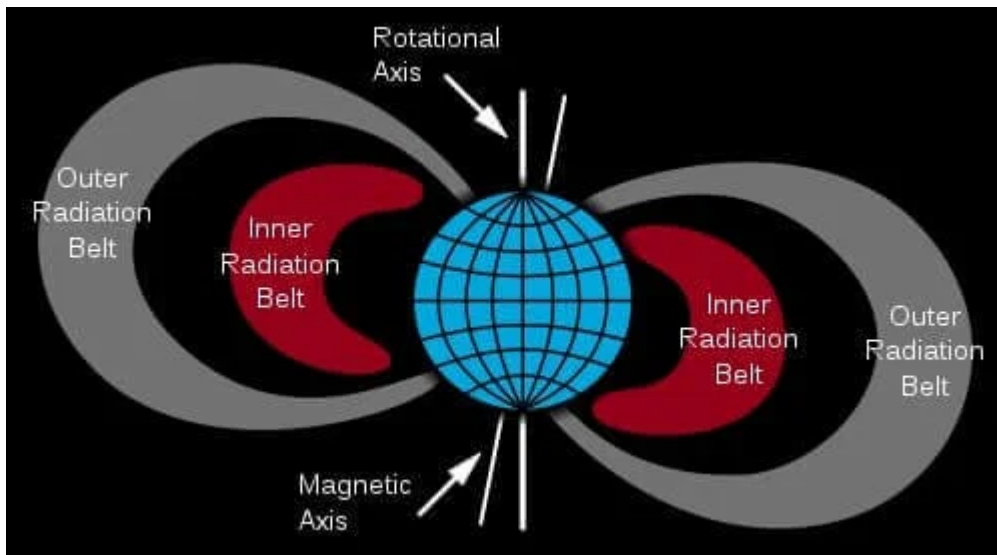


- The **ionosphere gets heated and distorted**, which means that **long-range radio communication that is dependent upon sub-ionospheric reflection can be difficult**.
- Ionospheric expansion can increase **satellite drag**, and it may become difficult to control their orbits.
- Geomagnetic storms disrupt satellite communication systems like GPS.
- Astronauts and high-altitude pilots would face high radiation levels.
- Electric power grids would see a high increase in voltage that would cause blackouts.
- Geomagnetic storms disrupt satellite communication systems like GPS.

Van Allen radiation belt

- A Van Allen radiation belt is a **zone of energetic charged particles**, most of which originate from the solar wind, that are **captured by and held around a planet by that planet's magnetic field**.
- There are two such concentric tire-shaped regions. The inner belt is 1–2 Earth radii out while the outer belt is at 4–7 Earth radii.
- By trapping the solar wind, the belts deflect the energetic particles and **protect the atmosphere**.
- The **belts endanger satellites**, which must have their sensitive components protected with adequate shielding if they spend significant time near that zone.
- Spacecraft travelling beyond low Earth orbit enter the zone of radiation of the Van Allen belts. Beyond the belts, they face additional hazards from cosmic rays and solar particle events.





Van Allen Belts (Booyabazooka, via [Wikimedia Commons](#))

Magnetic field of other solar system objects

Moon

- The magnetic field of the Moon is very weak in comparison to that of the Earth and doesn't have a magnetic dipole. It is not strong enough to prevent atmospheric stripping by the solar wind.

Mercury

- Mercury's magnetic field is approximately a magnetic dipole (meaning the field has two poles) and is just 1.1% that of Earth's magnetic field.
- Its proximity to the sun makes it next to impossible to sustain an atmosphere.

Mars

- Mars does not have an intrinsic global magnetic field, but the solar wind directly interacts with the atmosphere of Mars, leading to the formation of a magnetosphere.
- The **lack of a significant magnetosphere** is thought to be one reason for Mars's thin atmosphere.

Venus



- Its ionosphere separates the atmosphere from outer space and the solar wind.
- In spite of the absence of a magnetic field, **Venus's atmosphere** is one of the densest among the terrestrial planets.

Jupiter

- Jupiter has the **largest magnetic field** and a thick atmosphere.

Saturn

- Saturn's magnetosphere is the second largest of any planet in the Solar System after Jupiter.

Uranus and Neptune too have a significant and similar magnetic field.

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